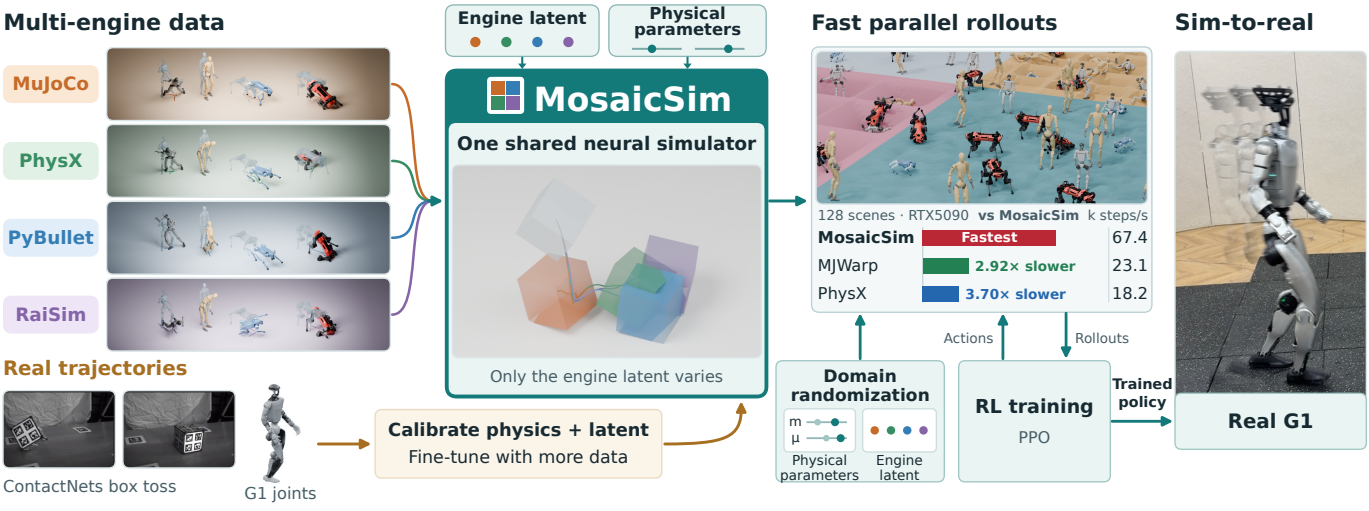

Multi-engine data
MuJoCo
PhysX
PyBullet
RaiSim
Real trajectories
ContactNets box toss
G1 joints
Engine latent
Physical parameters
MosaicSim
One shared neural simulator
Only the engine latent varies
Calibrate physics + latent
Fine-tune with more data
Fast parallel rollouts
128 scenes · RTX5090 vs MosaicSim k steps/s
MosaicSim Fastest 67.4
MJWarp 2.92× slower 23.1
PhysX 3.70× slower 18.2
Actions
Rollouts
Domain randomization
m
μ
Physical parameters
Engine latent
RL training
PPO
Trained policy
Sim-to-real
Real G1